Volatility Quant Fundamental Interview Questions

QuantInsider

Basic Volatility Concepts

1. **Define Historical (Realized) vs. Implied Volatility.**
   - What data do you need to compute each?
   - Under what circumstances might the two diverge?
2. **Explain why volatility is often quoted in "annualized" terms.**
   - How would you convert a daily realized-vol estimate into an annualized figure?
   - If you observe a 20% annualized vol, what is the approximate daily vol (assuming 252 trading days)?
3. **What is Volatility Skew (or Smile)?**
   - Sketch a typical equity-index vol smile vs. a typical FX vol smile.
   - What market-driven factors create a pronounced skew on equity options?
4. **Describe the difference between "at-the-money (ATM) volatility," "25**$\Delta risk-reversal, ̧and ̧25\Delta butte$

5. If a 25$\Delta risk-reversal is positive, what does that tell you about the implied skew? How would you compute a 25\Delta strangle$

Black–Scholes Greeks and Sensitivities

6. **Derive the Black–Scholes Vega formula.**
   - Provide the expression for vega of a European call or put.
   - Explain in words why vega peaks ATM and decays as you move deep ITM or OTM.
2. **Explain the concept of "Volatility Delta" (dVega/dSpot).**
   - How does a small move in spot (underlying price) affect implied vol when using an ATM-skewed surface?
3. **What is "Gamma Scalping"?**
   - Walk through step-by-step how buying a call and delta-hedging constitutes a gamma-scalp trade.
   - Clarify why gamma scalping alone is not the source of profit—what actually drives PL?

4. **Define and compare "Vanna" and "Volga/Charm."**
   - For a vol trader, why is vanna exposure important when managing a large Vega position?
   - If implied vol increases sharply when spot falls (i.e., skew moves), which greek kicks in and why?

Volatility Surfaces and Models

1. **What is a Local Volatility Model (Dupire)?**
   - State Dupire's forward PDE (local vol formula) in words.
   - How would you calibrate a local vol to a given set of market-quoted implied volatilities across strikes and tenors?
2. **Contrast Local Volatility vs. Stochastic Volatility.**
   - What are the main shortcomings of a pure local-vol framework in explaining the dynamics of the vol surface?
   - In what way does a simple one-factor stochastic-vol (e.g., Heston) model "patch" those shortcomings?
3. **Describe the Heston Model.**
   - Write down its SDEs under the risk-neutral measure.
   - Explain the roles of the parameters κ (mean reversion), θ (long-run variance), σ (vol-of-vol), ρ (correlation), and v_0 (initial variance).
   - What is the typical effect of increasing ρ (making it more negative) on the ATM skew?
4. **What is a SABR model?**
   - State the α, β, ρ, ν parameters and explain the "β" parameter's effect on forward vs. spot-dependent vol.
   - Describe how the SABR approximation (e.g., Hagan's formula) is used to generate an implied vol surface.
5. **Explain "Vol-of-Vol" (VVOL).**
   - How would you estimate VVOL from historical time series?
   - What is a "Volatility Swap," and how is its payoff related to realized variance?

Surface Dynamics and Calibration

1. **What does "sticky delta" vs. "sticky strike" mean in terms of implied vol surface dynamics?**
   - Give a scenario where sticky-delta dynamics are more realistic than sticky-strike.
2. **Walk through a simple-terms calibration routine for a Heston model.**
   - Suppose you have market-quoted implied vols at tenors 1M, 3M, and 6M for a set of strikes. How do you set up the optimization?

- What objective metric would you minimize, and why?

3. **How do you interpolate and extrapolate a discrete set of market-given implied vol points to construct a continuous surface?**
   - Mention common interpolation methods in strike (e.g., SVI, kriging, monotonic cubic splines) and in time (e.g., piecewise constant, SABR shape).
   - Why is arbitrage-free (no calendar or butterfly arbitrage) crucial when building the surface?

Term Structure and Calendar Spreads

1. **Define the term structure of volatility.**
   - How can one measure the forward-starting volatility between two future dates from the spot vol curve?
   - If 1M vol = 18% and 2M vol = 20%, what is the approximate implied 1M—2M forward vol?
2. **Explain a simple "Calendar Spread" vol trade.**
   - If the front-month implied vol is 25% and the next-month is 20%, how would you structure a vol calendar spread, and what directional view does it represent?
   - What risks are inherent in a calendar structure (e.g., vol-term-structure risk, gamma risk)?
3. **What is a "Dispersion Trade"?**
   - Explain how being short an index-implied vol and long a weighted average of constituent implied vols can be a neutral-delta, relative-value trade.
   - Under what market conditions would that trade tend to make money or lose money?

Econometric/Time-Series Approaches

1. **Describe a basic GARCH(1,1) model for forecasted variance.**
   - Write down the recursion for σ_t^2.
   - How would you backtest a GARCH forecast against realized variance?
2. **Explain the Exponentially Weighted Moving Average (EWMA) for variance.**
   - If you choose $\lambda = 0.94$, what fraction of emphasis do you put on today's squared return vs. yesterday's variance?
   - Compare the half-life ($\lambda \approx 0.94$) to a GARCH with estimated α, β.

Practical Vol Trading and Risk Management

1. **What is "Vol Carry"?**
   - How do you compute carry (in vol points) for rolling a 1M forward-start vol into the spot 1M by selling spot-1M and buying forward-2M, etc.?
   - Why might a vol-carry strategy fail during a volatility spike?

2. **Outline how you would hedge a vega-heavy book.**
   - What instruments (e.g., vanilla options, variance swaps, VIX futures) would you use?
   - How do you decide when to "pull in" some of your vega exposure versus letting it run?

3. **What is a "Variance Swap," and how is its fair strike computed?**
   - Provide the formula for the fair variance swap rate in terms of a strip of out-of-the-money options.
   - Explain why, in practice, the "log-contract" or "corridor-variance" approximation might deviate from realized volatility.

4. **Explain "VIX Futures" and "VIX Options."**
   - How is the VIX index itself constructed from SPX implied volatilities?
   - If you expect SPX realized vol to rise over the next month, how would you express that view via VIX futures or VIX calls?